CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 25: Probability

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Probability of an event

1. Which of these cannot be the probability of an event? (a) $\frac{5}{4}$ (b) 0.25 (c) $\frac{1}{33}$ (d) 67% ICSE 2025, 1 mark

SOLUTION

A probability always lies from 0 to 1. $\frac{5}{4} = 1.25$ is greater than 1.

Answer: (a)

2 Balls, numbered cards and letters

2. A box holds cards numbered 1 to 30, and one card is drawn at random. A wall clock is won for a perfect square, a water bottle for an even number that is also a multiple of 3, and a purse for a prime number. Find the probability of winning (a) the wall clock (b) the water bottle (c) the purse. ICSE 2025, 3 marks

SOLUTION

There are 30 equally likely outcomes.

(a) Perfect squares: 1, 4, 9, 16, 25 (5 cards). Probability $= \frac{5}{30} = \frac{1}{6}$.

(b) Even multiples of 3 are the multiples of 6: 6, 12, 18, 24, 30 (5 cards). Probability $= \frac{5}{30} = \frac{1}{6}$.

(c) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 (10 cards). Probability $= \frac{10}{30} = \frac{1}{3}$.

Answer: (a) $\frac{1}{6}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$

3. The letters A, D, M, N, O, S, U, Y are written on separate cards. One card is drawn at random. Find the probability that the letter (a) is in the word MONDAY (b) is not in MONDAY (c) is in both SUNDAY and MONDAY. ICSE 2024, 3 marks

SOLUTION

There are 8 cards.

(a) Letters of MONDAY: M, O, N, D, A, Y, which is 6 cards. Probability $= \frac{6}{8} = \frac{3}{4}$.

(b) $1 - \frac{3}{4} = \frac{1}{4}$ (the letters S and U).

(c) Common to both words: N, D, A, Y, which is 4 cards. Probability $= \frac{4}{8} = \frac{1}{2}$.

Answer: (a) $\frac{3}{4}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$

4. A letter is chosen at random from the English alphabet. Find the probability that it is a vowel. (a) $\frac{4}{26}$ (b) $\frac{5}{26}$ (c) $\frac{21}{26}$ (d) $\frac{5}{24}$ ICSE 2023, 1 mark

SOLUTION

There are 26 letters and 5 vowels (a, e, i, o, u).

Answer: (b)

5. A bag holds 25 cards numbered 1 to 25, and one card is drawn at random. Find the probability that its number is (a) a multiple of 5 (b) a perfect square (c) a prime number. ICSE 2023, 3 marks

SOLUTION

(a) Multiples of 5: 5, 10, 15, 20, 25 (5 cards). Probability $= \frac{5}{25} = \frac{1}{5}$.

(b) Perfect squares: 1, 4, 9, 16, 25 (5 cards). Probability $= \frac{1}{5}$.

(c) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23 (9 cards). Probability $= \frac{9}{25}$.

Answer: (a) $\frac{1}{5}$ (b) $\frac{1}{5}$ (c) $\frac{9}{25}$

6. Each letter of the word AUTHORIZES is written on a separate disc, and one disc is drawn at random. Find the probability that the letter is (a) a vowel (b) one of the first 9 letters of the alphabet (c) one of the last 9 letters of the alphabet. ICSE 2020, 3 marks

SOLUTION

The word has 10 different letters.

(a) Vowels: A, U, O, I, E (5 discs). Probability $= \frac{5}{10} = \frac{1}{2}$.

(b) The first 9 letters are A to I. In the word: A, H, I, E (4 discs). Probability $= \frac{4}{10} = \frac{2}{5}$.

(c) The last 9 letters are R to Z. In the word: U, T, R, Z, S (5 discs). Probability $= \frac{5}{10} = \frac{1}{2}$.

Answer: (a) $\frac{1}{2}$ (b) $\frac{2}{5}$ (c) $\frac{1}{2}$

7. A box holds 25 discs numbered 1 to 25, and one disc is drawn at random. Find the probability that its number is (a) odd (b) divisible by both 2 and 3 (c) less than 16. ICSE 2019, 3 marks

SOLUTION

(a) Odd numbers from 1 to 25: 13. Probability $= \frac{13}{25}$.

(b) Divisible by 2 and 3 means divisible by 6: 6, 12, 18, 24 (4 discs). Probability $= \frac{4}{25}$.

(c) Numbers less than 16: 1 to 15 (15 discs). Probability $= \frac{15}{25} = \frac{3}{5}$.

Answer: (a) $\frac{13}{25}$ (b) $\frac{4}{25}$ (c) $\frac{3}{5}$

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8. Cards numbered 2, 4, 6, 8, 10, 12, 14, 16, 18 and 20 are kept in a bag, and one is drawn at random. Find the probability that its number is (a) prime (b) divisible by 4 (c) a multiple of 6 (d) odd.

ICSE 2018, 4 marks

SOLUTION

There are 10 cards.

(a) The only prime is 2. Probability $= \frac{1}{10}$.

(b) Divisible by 4: 4, 8, 12, 16, 20. Probability $= \frac{5}{10} = \frac{1}{2}$.

(c) Multiples of 6: 6, 12, 18. Probability $= \frac{3}{10}$.

(d) There are no odd numbers. Probability $= 0$.

Answer: (a) $\frac{1}{10}$ (b) $\frac{1}{2}$ (c) $\frac{3}{10}$ (d) 0

9. Sixteen cards are labelled a, b, c, ..., o, p, and one is drawn at random. Find the probability that the letter is (a) a vowel (b) a consonant (c) not one of the letters of the word "median". ICSE 2017, 3 marks

SOLUTION

(a) Vowels among a to p: a, e, i, o (4 cards). Probability $= \frac{4}{16} = \frac{1}{4}$.

(b) $1 - \frac{1}{4} = \frac{3}{4}$

(c) The letters m, e, d, i, a, n are 6 of the 16 cards, so 10 cards are not among them. Probability $= \frac{10}{16} = \frac{5}{8}$.

Answer: (a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{5}{8}$

3 Coins and dice

10. Assertion (A): When a die is thrown once, the probability of getting an even number is $\frac{2}{3}$. Reason (R): The even numbers on a die are $\{2, 4, 6\}$. (a) A true, R false (b) A false, R true (c) both true (d) both false

ICSE 2024, 1 mark

SOLUTION

There are 3 even numbers out of 6, so the probability is $\frac{3}{6} = \frac{1}{2}$. A is false.

R is true.

Answer: (b)

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